

Green IT and Electronics: A Need of the Hour

Green information technology (IT) aims at designing and manufacturing such communication devices and electronic products that are environment friendly and support energy conservation. This article covers various aspects of green IT, including key components and recent findings, to create a better understanding among people.



In the present digital era, we are surrounded by several electronic devices such as mobile phones, computers, tablets, laptops, and other e-gadgets that form major part of our lives. These devices have revolutionised the way we work, spend our free time, or communicate with each other. This digital transformation has been possible only due to huge contributions by researchers, engineers, and scientists.

However, there is a big question that needs to be answered. Are we really ready for these changes? Do these changes have a negative impact on our life? The major drawback that can be inferred from this digital transformation is huge power

consumption and its implication on the environment. This particular drawback and its related effects have given rise to the concept of green computing, green IT or green electronics.

What is green IT?

Green information technology (IT) aims at minimising the negative intervention of IT operations or functionalities with the environment by designing, manufacturing, and using IT, electronics, or communication devices that are environment friendly and support energy conservation. Green IT not only deals with energy-saving or less power consumption but also focuses on e-waste reduction.

In 1992, this concept evolved when the US Environmental Protection Agency launched a voluntary labelling programme that helps organisations save money and reduce greenhouse gas emissions by identifying products that offer superior energy efficiency. Green computing is based on motives like conservation of energy and natural resources, reduction in usage of hazardous materials, recycling of e-waste, promotion of biodegradability, virtualisation, and usage of energy-efficient products (Figure 1).

Components of green IT

There are certain key components that form the base of green IT.

